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Innovative Applications of O.R.

Degradation-based maintenance decision using stochastic filtering for systems under imperfect maintenance

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ABSTRACT

The notion of imperfect maintenance has spawned a large body of literature, and many imperfect maintenance models have been developed. However, there is very little work on developing suitable imperfect maintenance models for systems outfitted with sensors. Motivated by the practical need of such imperfect maintenance models, the broad objective of this paper is to propose an imperfect maintenance model that is applicable to systems whose sensor information can be modeled by stochastic processes. The proposed imperfect maintenance model is founded on the intuition that maintenance actions will change the rate of deterioration of a system, and that each maintenance action should have a different degree of impact on the rate of deterioration. The corresponding parameter-estimation problem can be divided into two parts: the estimation of fixed model parameters and the estimation of the impact of each maintenance action on the rate of deterioration. The quasi-Monte Carlo method is utilized for estimating fixed model parameters, and the filtering technique is utilized for dynamically estimating the impact from each maintenance action. The competence and robustness of the developed methods are evidenced via simulated data, and the utility of the proposed imperfect maintenance model is revealed via a real data set.

1. Introduction

Maintenance actions can be classified, according to their efficiency, into three types: perfect maintenance, imperfect maintenance and minimal maintenance. The assumption of perfect maintenance is only applicable to structurally simple systems, and the assumption of minimal maintenance is only applicable to highly complex systems. By contrast, it is more realistic in true experience that maintenance actions are imperfect, merely restoring a maintained system's condition to somewhere between good-as-new and bad-as-old. To evaluate the impact of maintenance actions on a maintained system's condition, many imperfect maintenance models have been developed. One commonly used class of imperfect maintenance models is the virtual age model introduced by Kijima (1989). The virtual age model assumes that an imperfect repair reduces a maintained system's physical age (therein called virtual age) by an amount proportional to the physical age just before the maintenance, or by an amount proportional to the additional age accumulated since the last maintenance. Many researchers have used the concept of virtual age for modelling imperfect corrective maintenance and/or imperfect preventive maintenance; see, among others, Doyen and Gaudoin (2011), Bouguerra, Chelbi, and Rezg (2012), Dijoux and Idee (2013), Ramirez and Utne (2013), Ahmadi (2014) and Ramirez and Utne (2015). Another commonly used class of imperfect maintenance models is the improvement factor model introduced by Malik (1979). The improvement factor model assumes that an imperfect repair changes the time of the hazard rate curve to some newer time but not all the way to zero. Lin, Zuo, and Yam (2000) extended the improvement factor model, stating that an imperfect repair could change both the time and the slope of the hazard rate curve. The extended improvement factor model hence includes many documented imperfect maintenance models as special cases, e.g., the virtual age model. For recent research on the (extended) improvement factor model, the reader may refer to, e.g., Park, Chang, and Lie (2012), Xia, Xi, Zhou, and Du (2012) and Khatab, Ait-Kadi, and Rezg (2014). Other imperfect maintenance models that cannot be classified into the improvement factor model are the Brown–Proschan model (see, e.g., Doyen, 2011) and the geometric process model (see, e.g., Zhang, Xie, & Gaudoin, 2013). Reviewing works on imperfect maintenance models are given, e.g., by Wang and Pham (2006), Nakagawa (2006) and Shafiee and Chukova (2013).

We note that most of the documented work on imperfect maintenance is only concerned with age-based maintenance. When dealing with degradation-based maintenance, people typically adopt the assumption that maintenance actions are either minimal or perfect

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(see, e.g., Chen, Ye, Xiang, & Zhang, 2015). The issue of treating imperfect maintenance in the context of degradation-based maintenance has not received much attention and remains widely open. Zhou, Xi, and Lee (2007) employed the improvement factor model for modeling the impact of imperfect repairs on a continuously monitored deteriorating system. However, one of the major shortcomings of the improvement factor model is that it is only applicable to cases where the hazard rate function can be derived analytically. For many models, e.g. the non-stationary Wiener process, no analytical hazard rate function is available. Wang and Pham (2011) treated imperfect maintenance by lifting the degradation critical threshold proportionally. Apparently, their approach is confined to the realm of speculation. It is more desirable to develop an imperfect maintenance model by taking a physically meaningful approach. Instead of lifting the degradation critical threshold, some researchers, e.g. Nicolai, Frenk, and Dekker (2009), Van and Berenguer (2012) and Mercier and Castro (2013), treated imperfect maintenance by reducing the degradation level of a maintained system by a random amount. The uniform distribution is always employed for modeling the distribution of the random amount. We should point out that the degradation level of a maintained system before and after a repair can be exactly known (e.g., by virtue of sensors). Therefore, the amount of degradation reduced by the repair can be exactly known. Hence, the practicability of the random degradation-level-reduction paradigm is debatable. What maintenance technicians need is a practical and useful imperfect maintenance model for degradation-based maintenance. Motivated by the need of maintenance technicians and inspired by the extended improvement factor model, the broad objective of this paper is to propose an imperfect maintenance model, called random improvement factor model, for degradation-based maintenance.

The random improvement factor model is founded on the intuition that maintenance actions will change the rate of degradation of a system, other than the level of degradation. The degradation rate defined herein is the rate of accumulation of degradation (see, e.g., Meeker & Escobar, 1998). Repairs slowing down degradation rate can be easily visualized in the context of, e.g., coating operations. In order to slow down the deteriorating process, steel structures are usually protected by certain organic coating systems (Perrin, Merlatti, Aragon, & Margaillan, 2009). The degradation level after each coating operation may or may not be changed. Another type of maintenance is lubricating rotating gears, by which the wearing processes of the gears will slow down. The degradation level after lubrication remains unchanged. One distinguishing feature of the random improvement factor model is the introduction of a latent variable, taking into account the fact that each maintenance action should have a different degree of impact on the rate of degradation. Accordingly, the filtering technique is utilized for dynamically estimating each realization of the latent variable. For illustrative purpose, the Wiener process is employed due to its adaptability in modeling degradation phenomena. The Wiener process (in its many forms) has been applied in Meeker and Escobar (1998) to model the size of fatigue crack as a function of the number of cycles, in Wang (2010) to model the degradation of bridge beams due to chloride ion ingression, and in Ye, Wang, Tsui, and Pecht (2013) to model the wear of magnetic heads used in hard disk drives and the light output of a light emitting diode. Applications of the Wiener process are also reported in Nikulin, Limnios, Balakrishnan, Kahle, and Huber-Carol (2010), Bian and Gebraeel (2012), Son, Fouladirad, Barros, Levrat, and Iung (2013), Si, Chen, Wang, Hu, and Zhou (2013) and Si, Wang, Hu, and Zhou (2014), to name a few. We shall underscore that, after appropriate modifications, the following procedure can be readily applied to other stochastic processes.

The remainder of the paper is organized as follows. Section 2 is devoted to presenting the random improvement factor model (in Section 2.1) and a generic maintenance scheme (in Section 2.2). Section 3 is devoted to real-time updating the degradation rate function of the maintained system (in Section 3.1) and the estimates of other fixed model parameters (in Section 3.2). Section 4 gives numerical examples to show the applicability and competence of the advanced techniques. Section 5 outlines concluding remarks and future work.

2. Model formulation

2.1. Random improvement factor model

Let $v(t)$ be a differentiable, non-negative, real-valued function of $t(\geq 0)$ with $v(0) = 0$. A Wiener process, denoted by $\{X_t, t \geq 0\}$, with drift function $v(t)$ and variance coefficient $\sigma(>0)$ is defined by $X_t = v(t) + \sigma W_t$; see, e.g., Si, Wang, Hu, Zhou, and Pecht (2012), Bian and Gebraeel (2012) and Son et al. (2013). $\{W_t, t \geq 0\}$ is the standard Brownian motion. The Wiener process has independent and normally distributed random increments. That is, for all $0 \leq s < t$, $X_t - X_s$ is independent of X_s and has the normal distribution $N(v(t) - v(s), \sigma^2(t - s))$. If $v(t)$ is a linear function, then $\{X_t, t \geq 0\}$ is called a stationary Wiener process. If $v(t)$ is a non-linear function, then $\{X_t, t \geq 0\}$ is called a non-stationary Wiener process. Suppose that a system, whose degradation conforms to the Wiener process $\{X_t, t \geq 0\}$, is put into operation at time 0. The expected degradation level of the system up to any time t is $\mathrm{E}[X_t] = v(t)$. Therefore, the first-order derivative of $v(t)$, denoted by $v'(t)$, can be treated like the degradation rate function of the underlying degradation process. After being put into operation at time 0, the system is repaired at time $s(>0)$. Right before the repair the degradation rate is $v'(s)$. Assume that the repair takes negligible time. The random improvement factor model says that right after the repair the degradation rate will be $bv'(s)$ with $0 < b < 1$. Here, b is a degradation-rate-reduction factor. It is noteworthy that each repair should have a different degree of impact on the degradation rate function. Therefore, the random improvement factor model says that the degradation-rate-reduction factor, b, is a random variable having certain distribution. Moreover, b is indeed a latent variable because the effect of a repair can only be inferred. From time s forward, the degradation rate function will be re-written by $bv'(s + t)$ for $t > 0$. Accordingly, from time s forward, the drift function will be re-written by $bv(s + t) + c$ for $t > 0$. Mathematically, the drift function is an integral of the degradation rate function. Hence, c can be treated like the constant of integration. Practically, c is introduced due to the fact that the repair may or may not shift the degradation level. Hence, we might call c a shifting factor.

Remark 1. Like the degradation-rate-reduction factor, the shifting factor should differ for different repairs. Though c is not fixed, it can be uniquely derived, given that all the other parameters are known (estimated). Suppose that a repair is performed at time $t_1(>0)$. The degradation level right before the repair is $x_{t_1^-} = v(t_1) + \sigma w_{t_1}$. w_{t_1} is a realization of W_{t_1}. The degradation level right after the repair is $x_{t_1^+} = b_1 v(t_1) + c_1 + \sigma w_{t_1}$. b_1 and c_1 are respectively the degradation-rate-reduction factor and shifting factor produced by the repair. The degradation measurement $x_{t_1^+}$ can equal or differ from the degradation measurement $x_{t_1^-}$, depending on the value of c_1. Once $x_{t_1^-}$ and $x_{t_1^+}$ are obtained, the shifting factor c_1 can be uniquely derived: $c_1 = x_{t_1^+} - x_{t_1^-} + v(t_1) - b_1 v(t_1)$. Therefore, every shifting factor can be evaluated till the end, when all the other unknown parameters (e.g., b_1) are estimated. Hence, in Section 3, we only developed methods for evaluating the other model parameters.

Remark 2. From time t_1 forward, the underlying degradation process can be modeled by $\{X_t = b_1 v(t) - b_1 v(t_1) + x_{t_1^+} + \sigma W_{t-t_1}, t > t_1\}$ which involves no shifting factor. Suppose that another repair is performed at time $t_2(> t_1)$. The degradation level at time t_2, right before the second repair, is $x_{t_2^-} = b_1 v(t_2) - b_1 v(t_1) + x_{t_1^+} + \sigma w_{t_2 - t_1}$. $w_{t_2-t_1}$ is a realization of $W_{t_2-t_1}$. The second degradation increment is

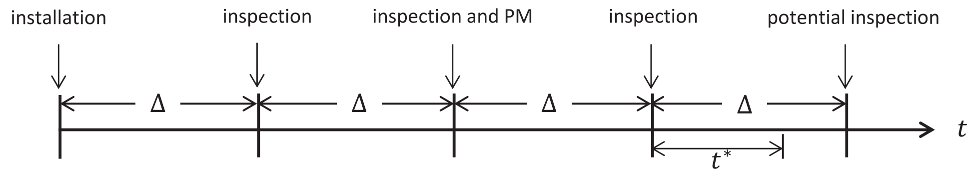

Fig. 1. A possible course of a replacement cycle.

$x_{t_2^-} - x_{t_1^+} = b_1 v(t_2) - b_1 v(t_1) + \sigma w_{t_2 - t_1}$. Note that all the other model parameters depend only on the degradation increments $x_{t_1^-}$ and $x_{t_2^-} - x_{t_1^+}$ not on the amount of shifted degradation $x_{t_1^+} - x_{t_1^-}$. Therefore, when estimating the other parameters, we might assume that repairs do not change the degradation level. This assumption is made only for ease of exposition: without introducing too many subscripts or superscripts.

It can be seen that the random improvement factor model is analogous to the improvement factor model except that (1) the random improvement factor model proceeds from the degradation rate point of view while the improvement factor model proceeds from the hazard rate point of view; (2) the degradation-rate-reduction factor is random while, for example, the age-reduction factor in the improvement factor model is identical for all repairs. In terms of degradation-based maintenance, treating imperfect maintenance via the degradation rate function instead of the hazard rate function has two main advantages. Firstly, deriving the hazard rate function via the first hitting time distribution function is mathematically intractable, especially when the degradation process is non-stationary. Secondly, unlike the hazard rate which is unobservable and unmeasurable, the impact of repairs on the degradation rate can be exteriorized from consecutive degradation measurements. The random improvement factor model can be generalized to include another (random) factor for reducing the physical age of a maintained system. This topic is beyond the scope of this paper.

2.2. A maintenance scheme

A generic maintenance scheme is developed for elucidating our methods. The degradation of a system, after its installation at time 0, will be monitored at time points $\{i\triangle : i \geq 1, \triangle > 0\}$. The degradation-monitoring (DM) points need not be equally spaced. Generalization of the maintenance scheme to irregular DM points is a straightforward matter. For $i \geq 1$, let x_i denote the degradation of the system monitored at the ith DM point $i\triangle$. After obtaining x_i, the degradation rate function of the system will be immediately updated by utilizing the filtering technique. After updating the degradation rate function, we need to take one of the following maintenance actions in order to optimize the maintenance scheme.

- Monitor the degradation of the system at the next DM point $(i+1)\triangle$.
- Preventively repair the system at time $i\triangle$ and monitor the degradation of the system at the next DM point $(i+1)\triangle$.
- Preventively replace the system at a time point prior to the next DM point $(i+1)\triangle$.
- Preventively repair the system at time $i\triangle$ and preventively replace the system at a time point prior to the next DM point $(i+1)\triangle$.

If the degradation of the system is monitored at the next DM point, the foregoing procedure repeats. If the system is preventively replaced at a time point prior to the next DM point, the foregoing procedure renews. If the system fails unexpectedly, it will be immediately replaced by a new physically and statistically identical one. We make the following mild assumptions on the maintenance scheme.

- Preventive maintenance (PM) is imperfect. Replacement, either preventive or unexpected, is perfect.
- Monitoring is perfect in the sense that it reveals the true degradation level of the system and does not change the condition of the system.
- Preventive maintenance actions take negligible time compared to the expected lifetime of the maintained system.

The cost of each inspection is C_I. The average costs of a PM action and a replacement are C_p and C_r, respectively. Unexpected failures can be detected instantaneously, each costing C_f. Practical conditions define the constraints of the maintenance costs as follows: $0 < C_I < C_p < C_r < C_f$.

Fig. 1 demonstrates a possible course of a replacement cycle, i.e. the time from the installation of a system to its replacement. At the first DM point, $\triangle$, after obtaining a degradation datum and updating the degradation rate function, the optimal maintenance action is to subject the system to the next inspection with no PM action taken at time $\triangle$. At the second DM point, $2\triangle$, after obtaining another degradation datum and updating the degradation rate function, the optimal maintenance decision is to subject the system to the next inspection with a PM action taken at time $2\triangle$. At the third DM point, the optimal maintenance decision is to preventively replace the system at time $3\triangle + t^*$ with no PM action taken at time $3\triangle$. If the system fails unexpectedly (namely, before time $3\triangle + t^*$), it will be immediately replaced by a new one.

The optimization of the maintenance scheme depends on the minimization of the long-run cost per unit time. As with convention, the lifetime of the system is defined as the first hitting time of the degradation of the system to a pre-determined critical threshold $l(>0)$. Let T_l denote the lifetime of the system. Given that the maintained system is still operating at the ith DM point, we introduce the following notation.

- $\ddot{X}_i$: the set of degradation-monitoring data up to and including the ith, $\ddot{X}_i = \{x_1, x_2, \ldots, x_i\}$.
- $\ddot{M}_i$: the set of maintenance data up to and including the ith, $\ddot{M}_i = \{m_1 m_2, \ldots, m_i\}$. If the system is preventively maintained at the ith DM point, we set $m_i = 1$. If no PM action is taken at the ith DM point, we set $m_i = 0$.
- $EC^i_{pm}(t)$: the expected cost rate of a replacement cycle with a PM action taken at the ith DM point and the system preventively replaced at time $i\triangle + t$.
- $EC^i_{na}(t)$: the expected cost rate of a replacement cycle with no repair at the ith DM point and the system preventively replaced at time $i\triangle + t$.

Let t_p^* denote the optimal preventive-replacement time that minimizes the cost rate function $EC^i_{pm}(t)$. Let t_n^* denote the optimal preventive-replacement time that minimizes the cost rate function $EC^i_{na}(t)$. According to the renewal theorem, the long-run cost per unit time is equal to the expected cost rate of a renewal cycle. Hence, to minimize the long-run cost per unit time, we only need to minimize the expected cost rate of a renewal cycle. Therefore,

- if $EC^i_{pm}(t_p^*) < EC^i_{na}(t_n^*)$ and $t_p^* < \triangle$, the system will be repaired at time $i\triangle$ and preventively replaced at time $i\triangle + t_p^*$;
- if $EC^i_{pm}(t_p^*) < EC^i_{na}(t_n^*)$ and $t_p^* > \triangle$, the system will be repaired at time $i\triangle$ and inspected at the next DM point $(i+1)\triangle$;

- if $EC^i_{pm}(t^*_p) > EC^i_{na}(t^*_n)$ and $t^*_n < \triangle$, the system will not be repaired at time $i\triangle$ but preventively replaced at time $i\triangle + t^*_n$;
- if $EC^i_{pm}(t^*_p) > EC^i_{na}(t^*_n)$ and $t^*_n > \triangle$, the system will not be repaired at time $i\triangle$ but inspected at the next DM point $(i+1)\triangle$.

The expected cost rate functions, $EC^i_{pm}(t)$ and $EC^i_{na}(t)$, can be formulated in the following manner. Given degradation data $\ddot{X}_i$ and maintenance data $\ddot{M}_i$, let $P_{rul}(t|\ddot{X}_i, \ddot{M}_i)$ denote the remaining useful life distribution function of the system:

$$P_{rul}(t|\ddot{X}_i, \ddot{M}_i) = \Pr(T_l \le t + i\triangle|\ddot{X}_i, \ddot{M}_i).$$

If the system is preventively repaired at time $i\triangle$ and will be preventively replaced at time $i\triangle + t$, the expected cost of the replacement cycle can be formulated to be

$$\begin{aligned}\mathrm{E}[\mathrm{cost}|\ddot{X}_i, \ddot{M}_{i-1}, m_i = 1] &= ic_I + \sum_{j=1}^{i} m_j c_p \\ &+ c_r[1 - P_{rul}(t|\ddot{X}_i, \ddot{M}_{i-1}, m_i = 1)] + (c_f + c_r)P_{rul}(t|\ddot{X}_i, \ddot{M}_{i-1}, m_i = 1).\end{aligned}$$

Here, ic_I is the total inspection cost. $\sum_{j=1}^{i} m_j c_p$ is the total PM cost. $c_r[1 - P_{rul}(t|\ddot{X}_i, \ddot{M}_i)]$ is the expected cost of a preventive replacement at time $i\triangle + t$. $(c_f + c_r)P_{rul}(t|\ddot{X}_i, \ddot{M}_i)$ is the expected cost of an unexpected failure before time $i\triangle + t$. The expected length of the replacement cycle can be formulated to be

$$\begin{aligned}&\mathrm{E}[\mathrm{length}|\ddot{X}_i, \ddot{M}_{i-1}, m_i = 1] \\ &= i\triangle + \int_0^t s \Pr(T_l = s + i\triangle|\ddot{X}_i, \ddot{M}_{i-1}, m_i = 1)ds \\ &\quad + t \Pr(T_l > t + i\triangle|\ddot{X}_i, \ddot{M}_{i-1}, m_i = 1),\end{aligned}$$

which can be further simplified to be

$$\begin{aligned}&\mathrm{E}[\mathrm{length}|\ddot{X}_i, \ddot{M}_{i-1}, m_i = 1] \\ &= i\triangle + \int_0^t s\, dP_{rul}(s|\ddot{X}_i, \ddot{M}_{i-1}, m_i = 1) + t[1 - P_{rul}(t|\ddot{X}_i, \ddot{M}_{i-1}, m_i = 1)].\end{aligned}$$

Therefore, the expected cost rate function $EC^i_{pm}(t)$ can be formulated to be

$$\begin{aligned}EC^i_{pm}(t) &= \frac{\mathrm{E}[\mathrm{cost}|\ddot{X}_i, \ddot{M}_{i-1}, m_i = 1]}{\mathrm{E}[\mathrm{length}|\ddot{X}_i, \ddot{M}_{i-1}, m_i = 1]} \\ &= \frac{ic_I + \sum_{j=1}^{i} m_j c_p + c_r + c_f P_{rul}(t|\ddot{X}_i, \ddot{M}_{i-1}, m_i = 1)}{i\triangle + t - \int_0^t P_{rul}(s|\ddot{X}_i, \ddot{M}_{i-1}, m_i = 1)ds}.\end{aligned}$$

By analogy, given no repair at time $i\triangle$ and a scheduled preventive replacement at time $i\triangle + t$, we have

$$EC^i_{na}(t) = \frac{ic_I + \sum_{j=1}^{i} m_j c_p + c_r + c_f P_{rul}(t|\ddot{X}_i, \ddot{M}_{i-1}, m_i = 0)}{i\triangle + t - \int_0^t P_{rul}(s|\ddot{X}_i, \ddot{M}_{i-1}, m_i = 0)ds}.$$

3. Parameter estimation

For the practical implementation of the random improvement factor model, the fundamental problem is how to estimate unknown parameters. We note that the degradation-rate-reduction factor is a latent random variable. By contrary, the other unknown model parameters (e.g., σ) are fixed. Therefore, different methods are developed for evaluating different types of model parameters.

3.1. Degradation-rate-reduction factor

A commonly used formulation of the drift function is given by $v(t) = \lambda t^\theta$ (see, e.g., Wang, Hu, Wang, & Si, 2014). $\lambda(> 0)$ is a scale parameter. $\theta(>0)$ is a shape parameter. The degradation rate function hence can be written by $v'(t) = \lambda\theta t^{\theta-1}$. The degradation-rate-reduction factor is assumed to follow a beta distribution:

$$\Pr(b = z|u, v) = \frac{\Gamma(u+v)}{\Gamma(u)\Gamma(v)} z^{u-1}(1-z)^{v-1}, \quad 0 < z < 1.$$

$u(>0)$ and $v(>0)$ are two unknown shape parameters. Let Beta(u, v) denote the beta distribution. The beta density function can take a wide variety of different shapes depending on the values of the two parameters. The mean and variance of the beta distribution are

$$\mathrm{E}[b] = \frac{u}{u+v} \quad \text{and} \quad \mathrm{Var}(b) = \frac{uv}{(u+v)^2(u+v+1)}.$$

For any $i \ge 1$, if a maintenance action is performed at time $i\triangle$, let b_i denote the degradation-rate-reduction factor produced by the repair. Immediately after the maintenance, the degradation rate function will be $\prod_{\substack{j=1,\dots,i \\ m_j=1}} b_j \lambda\theta t^{\theta-1}$ with $t > i\triangle$. Let λ_i denote $\prod_{\substack{j=1,\dots,i \\ m_j=1}} b_j\lambda$ and λ_0 denote λ. The degradation rate function hence can be simplified to be $\lambda_i\theta t^{\theta-1}$. Instead of saying that after the repair the degradation rate function changes from $\prod_{\substack{j=1,\dots,i-1 \\ m_j=1}} b_j\lambda\theta t^{\theta-1}$ to $\prod_{\substack{j=1,\dots,i \\ m_j=1}} b_j\lambda\theta t^{\theta-1}$, we might say that after the repair the degradation rate function changes from $\lambda_{i-1}\theta t^{\theta-1}$ to $\lambda_i\theta t^{\theta-1}$. That is, equivalently, the scale parameter of the degradation rate function changes from λ_{i-1} to λ_i. Therefore, to update the degradation rate function, we only need to update the estimate of the scale parameter.

If no repair is performed at time $i\triangle$, then we have $\lambda_i = \lambda_{i-1}$. If the system is preventively repaired at time $i\triangle$, we have $\lambda_i = b_i\lambda_{i-1}$, which can be further written into

$$\lambda_i = \frac{u}{u+v}\lambda_{i-1} + \left(b_i - \frac{u}{u+v}\right)\lambda_{i-1}.$$

The relation between λ_i and λ_{i-1} can be summarized into

$$\lambda_i = \mu_i\lambda_{i-1} + \delta_i, \tag{1}$$

in which

$$\mu_i = \left(\frac{u}{u+v}\right)^{m_i} \quad \text{and} \quad \delta_i = \left(b_i - \frac{u}{u+v}\right)\lambda_{i-1}m_i.$$

Let Q_i denote the variance of δ_i. We have

$$Q_i = \frac{uv\lambda_{i-1}^2}{(u+v)^2(u+v+1)}m_i.$$

From time $i\triangle$ forward, the degradation of the system can be modeled by $\{X_t = \lambda_i t^\theta - \lambda_i(i\triangle)^\theta + x_i + \sigma W_{t-i\triangle}, t > i\triangle\}$. By definition, for any $t(> i\triangle)$, the random increment $X_t - X_{i\triangle}$ follows the normal distribution $N(\lambda_i t^\theta - \lambda_i(i\triangle)^\theta, \sigma^2(t - i\triangle))$. Therefore, the sample $x_{i+1} - x_i$ shall come from the normal distribution $N(\lambda_i[(i+1)\triangle]^\theta - \lambda_i(i\triangle)^\theta, \sigma^2\triangle)$. That is, we have

$$x_{i+1} - x_i = \lambda_i[(i+1)\triangle]^\theta - \lambda_i(i\triangle)^\theta + \omega_{i+1},$$

in which ω_{i+1} is a random sample from the normal distribution $N(0, \sigma^2\triangle)$. Let ρ_{i+1} denote $[(i+1)\triangle]^\theta - (i\triangle)^\theta$ and y_{i+1} denote $x_{i+1} - x_i$. The relation between x_{i+1} and x_i can be summarized into

$$y_{i+1} = \rho_{i+1}\lambda_i + \omega_{i+1}. \tag{2}$$

Apparently, Eqs. (1) and (2) construct a linear dynamic system. Particularly, Eq. (1) is a process equation, and δ_i is the zero-mean non-Gaussian process noise having variance Q_i; Eq. (2) is a measurement equation, and ω_{i+1} is the zero-mean Gaussian measurement noise having variance $\sigma^2\triangle$. Therefore, the Kalman filter can be utilized for real-time estimating λ_i. The Kalman filter is the minimum-variance state estimator for linear dynamic systems with Gaussian noise (Rhodes, 1971). In addition, the Kalman filter is the minimum-variance state estimator for linear dynamic systems with non-Gaussian noise (Simon, 2006). Applications of the Kalman filter and its modifications have been extensively reported in many fields including statistical signal processing, economics, biostatistics and engineering. A representative sample of reviewing works on the Kalman filter and its modifications includes Voss, Timmer, and Kurths (2004), Cappe, Godsill, and Moulines (2007) and Khatibisepehr, Huang, and Khare (2013).

For any $i \geq 1$, let $\hat{\lambda}_i^-$ denote the a priori estimate of λ_i, calculated by utilizing the degradation data $\ddot{X}_i$ and maintenance data $\ddot{M}_i$. Let $\hat{\lambda}_i$ denote the a posteriori estimate of λ_i, calculated by utilizing the degradation increment y_{i+1} and the a priori estimate $\hat{\lambda}_i^-$. Define P_i^- to be the a priori estimate error variance:

$$P_i^- = \mathrm{E}[(\lambda_i - \hat{\lambda}_i^-)(\lambda_i - \hat{\lambda}_i^-)].$$

Define P_i to be the a posteriori estimate error variance:

$$P_i = \mathrm{E}[(\lambda_i - \hat{\lambda}_i)(\lambda_i - \hat{\lambda}_i)].$$

The recursive estimation of the scale parameter entails the following updating equations (Simon, 2006):

- state estimate propagation

$$\hat{\lambda}_i^- = \mu_i \hat{\lambda}_{i-1},$$

- estimate error variance propagation

$$P_i^- = \mu_i^2 P_{i-1} + Q_i,$$

- Kalman gain

$$G_i = \frac{P_i^- \rho_{i+1}}{P_i^- \rho_{i+1}^2 + \sigma^2 \triangle},$$

- state estimate update

$$\hat{\lambda}_i = \hat{\lambda}_i^- + G_i (y_{i+1} - \rho_{i+1} \hat{\lambda}_i^-),$$

- estimate error variance update

$$P_i = (1 - G_i \rho_{i+1}) P_i^-.$$

Therefore, the a posteriori estimate of λ_i can be updated by equation

$$\hat{\lambda}_i = \mu_i \hat{\lambda}_{i-1} + \frac{(\mu_i^2 P_{i-1} + Q_i)\, \rho_{i+1}}{(\mu_i^2 P_{i-1} + Q_i)\, \rho_{i+1}^2 + \sigma^2 \triangle} (y_{i+1} - \rho_{i+1} \mu_i \hat{\lambda}_{i-1}),$$

and the a posteriori estimate error variance can be updated by equation

$$P_i = \frac{(\mu_i^2 P_{i-1} + Q_i)\, \sigma^2 \triangle}{(\mu_i^2 P_{i-1} + Q_i)\, \rho_{i+1}^2 + \sigma^2 \triangle}.$$

The starting values $\hat{\lambda}_0$ and P_0 are chosen arbitrarily, reflecting our best assessment of the scale parameter λ. It is worth stressing that the performance of the above Kalman filter depends on parameters $\{\theta, \sigma, u, v\}$ but not on λ. Given a rough estimate of λ, i.e. $\hat{\lambda}_0$, the a posteriori estimate $\hat{\lambda}_i$ will always converge to λ_i with i increasing, provided that the parameters $\{\theta, \sigma, u, v\}$ are accurately estimated. After updating the degradation rate function, we need to update the remaining useful life distribution function $P_{rul}(t|\ddot{X}_i, \ddot{M}_i)$. Appendix A develops an approximation to the remaining useful life distribution function.

3.2. Other fixed parameters

The other parameters are fixed and can be evaluated by using historical degradation data and maintenance data that have been collected. Suppose that we have collected degradation data $\ddot{X}_n = \{x_1, x_2, \ldots, x_n\}$ and maintenance data $\ddot{M}_{n-1} = \{m_1, m_2, \ldots, m_{n-1}\}$ in which $n (\geq 1)$ is the sample size. Let M_n denote the total number of repairs: $M_n = \sum_{j=1}^{n-1} m_j$. Define a vector $\boldsymbol{b}_{M_n}$, with length M_n, recording the M_n degradation-rate-reduction factors produced by the M_n PM actions. For example, if maintenance actions are performed only at the second and fifth DM points ($5 < n$), then we have $M_n = 2$ and $\boldsymbol{b}_{M_n} = (b_2, b_5)$. Note that if a maintenance action is performed, the realization of the degradation-rate-reduction factor, on which the likelihood function is conditioned, is latent. Let Θ denote the vector of fixed model parameters: $\Theta = (\lambda, \theta, \sigma, \mu, v)$. Given degradation data $\ddot{X}_n$, a general formulation of the likelihood function of Θ is given by

$$\begin{aligned}\mathfrak{L}(\Theta|\ddot{X}_n) = & \int_0^1 \cdots \int_0^1 \frac{1}{(\sigma\sqrt{2\pi\triangle})^n} \\ & \times \exp\left(-\frac{(x_1 - \lambda\rho_1)^2}{2\sigma^2\triangle} - \frac{1}{2\sigma^2\triangle}\sum_{k=2}^{n}\left(y_k - \prod_{\substack{j=1,\ldots,k-1 \\ m_j=1}} b_j \lambda \rho_k\right)^2\right) \\ & \times \left[\frac{\Gamma(u+v)}{\Gamma(u)\Gamma(v)}\right]^{M_n} \prod_{\substack{j=1,\ldots,n-1 \\ m_j=1}} b_j^{u-1}(1-b_j)^{v-1} d\boldsymbol{b}_{M_n}.\end{aligned}$$

The maximum likelihood estimate of Θ, denoted by $\hat{\Theta}$, is the maximizer of the likelihood function:

$$\hat{\Theta} = \arg\max_{\Theta} \mathfrak{L}(\Theta|\ddot{X}_n).$$

Normally, the likelihood function $\mathfrak{L}(\Theta|\ddot{X}_n)$ shall involve high-dimensional integration and therefore is very knotty to evaluate. Two available approaches to tackling high-dimensional integration are numerical integration and Monte Carlo integration. One drawback of numerical integration is that it is inappropriate for evaluating the likelihood when $M_n > 4$, due to the curse of dimensionality. Monte Carlo integration, on the contrary, is particularly useful for evaluating high-dimensional integrals.

Let non-negative function $g(\boldsymbol{b}_{M_n}; \Theta)$ denote the integrant:

$$\begin{aligned}g(\boldsymbol{b}_{M_n}; \Theta) = & \frac{1}{(\sigma\sqrt{2\pi\triangle})^n} \\ & \times \exp\left(-\frac{(x_1 - \lambda\rho_1)^2}{2\sigma^2\triangle} - \frac{1}{2\sigma^2\triangle}\sum_{k=2}^{n}\left(y_k - \prod_{\substack{j=1,\ldots,k-1 \\ m_j=1}} b_j \lambda \rho_k\right)^2\right) \\ & \times \left[\frac{\Gamma(u+v)}{\Gamma(u)\Gamma(v)}\right]^{M_n} \prod_{\substack{j=1,\ldots,n-1 \\ m_j=1}} b_j^{u-1}(1-b_j)^{v-1}.\end{aligned}$$

The likelihood function hence can be simplified to be

$$\mathfrak{L}(\Theta|\ddot{X}_n) = \int_{\boldsymbol{U}_{M_n}} g(\boldsymbol{b}_{M_n}; \Theta) d\boldsymbol{b}_{M_n},$$

in which $\boldsymbol{U}_{M_n}$ is the M_n-dimensional unit cube: $\boldsymbol{U}_{M_n} = \underbrace{(0,1) \times \cdots \times (0,1)}_{M_n}$. The idea of Monte Carlo integration is to select at random N points, denoted by $\boldsymbol{u}_1, \ldots, \boldsymbol{u}_N$, from the unit cube $\boldsymbol{U}_{M_n}$ by performing N independent trials. Then the Monte Carlo approximation for the likelihood is

$$\mathfrak{L}(\Theta|\ddot{X}_n) \approx \frac{1}{N} \sum_{k=1}^{N} g(\boldsymbol{u}_k; \Theta).$$

The strong law of large numbers guarantees that the Monte Carlo approximation converges almost surely. Moreover, it follows from the central limit theorem that the expected integration error is $O(N^{-1/2})$ (Gentle, 2003).

The main disadvantage of Monte Carlo integration is that it is slow. N should be fairly large – of the order of thousands or even millions – to obtain acceptable precision in the answer. On the other hand, we note that optimization algorithms for maximizing $\mathfrak{L}(\Theta|\ddot{X}_n)$ shall require thousands of times of integration. Hence, to reduce computational load, we here replace Monte Carlo integration with quasi-Monte Carlo integration (Morokoff & Caflisch, 1995). It has been proved that lower error and improved convergence can be obtained by replacing random points with more uniformly distributed (known as quasi-random) points; see Niederreiter (1978). Computational experiments have shown that quasi-Monte Carlo integration is significantly more efficient than Monte Carlo integration; see, among others, Papageorgiou (2001), Schlier (2004), and Sloan and Wozniakowski (1998). In Section 4, Halton sequence (for dimensions up to 6) and Sobol sequence (for dimensions more than 6) are employed to generate quasi-random points from the unit cube $\boldsymbol{U}_{M_n}$. Differential evolution (Storn & Price, 1997) is utilized for global optimization. For later reference, we might call the foregoing parameter-estimation method by utilizing quasi-Monte Carlo integration the "QMC method".

4. Numerical examples

The competence and robustness of the Kalman filter and QMC method are evidenced via simulated data, and the utility of the random improvement factor model is revealed via a real data set. Simulated degradation data are generated via the following method, called DG method. At time $\triangle$, we randomly simulate a degradation datum, denoted by x_1, from the normal distribution $N(\lambda\triangle^{\theta}, \sigma^2\triangle)$.

- If an imperfect repair is then performed at time $\triangle$, the value of the scale parameter changes, immediately after the repair, from λ to $b_1\lambda$. b_1 is randomly simulated from the beta distribution $\text{Beta}(u, v)$. At time $2\triangle$, we randomly simulate a degradation increment, denoted by y_2, from the normal distribution $N(b_1\lambda(2^{\theta} - 1)\triangle^{\theta}, \sigma^2\triangle)$.
- If no repair is performed at time $\triangle$, at time $2\triangle$ we randomly simulate a degradation increment, y_2, from the normal distribution $N(\lambda(2^{\theta} - 1)\triangle^{\theta}, \sigma^2\triangle)$.

The degradation datum obtained at time $2\triangle$ is denoted by x_2: $x_2 = x_1 + y_2$. By analogy, at time $i\triangle$ $(i \geq 3)$, we randomly simulate a degradation increment, denoted by y_i, from the normal distribution $N(\prod_{\substack{j=1,\ldots,i-1 \\ m_j=1}} b_j\lambda[i^{\theta} - (i-1)^{\theta}]\triangle^{\theta}, \sigma^2\triangle)$. The ith degradation datum is denoted by x_i: $x_i = x_{i-1} + y_i$. $\{b_i | m_i = 1, i \geq 1\}$ are all randomly simulated from the beta distribution $\text{Beta}(u, v)$. $\{m_i, i \geq 1\}$ are the maintenance data defined in Section 2.

4.1. Performance of the Kalman filter

Simulated degradation data, generated by the DG method, are used to serve the purpose. The true values of the fixed model parameters are set to be $\lambda = 7\text{e}{-5}$, $\theta = 2.5$, $\sigma = 2$, $u = 12$ and $v = 4$. Hence the mean and standard deviation of the degradation-rate-reduction factor are 0.75 and 0.11, respectively. The corresponding probability density function of the degradation-rate-reduction factor is plotted in Fig. 2. It can be seen that the probability density function is negatively skewed with its mode around 0.78. The length of the DM interval is 100 units of time, i.e., $\triangle = 100$. The maintenance data $\{m_i, i \geq 1\}$ are randomly generated from the Bernoulli distribution with the success probability being 0.5. That is, the probability of taking a repair at a DM point is 0.5. The starting values for the Kalman filter are set to be $\hat{\lambda}_0 = 1$ and $P_0 = 1$, which are far away from the true values.

Run the recursive filter on simulated degradation data to produce real-time estimates of the scale parameter. To inquire into the performance of the Kalman filter, we plot the sequence $\{\lambda_{i-1}, i \geq 1\}$ and the sequence $\{\hat{\lambda}_{i-1}, i \geq 1\}$ in Fig. 3. It is clear from Fig. 3 that the estimate $\hat{\lambda}_{i-1}$ converges to the true value λ_{i-1} very quickly, even though the starting value $\hat{\lambda}_0$ is far away from the true value λ. After obtaining 10 degradation measurements, the filter is able to produce a fairly accurate estimate. At each DM point $i\triangle$ $(i \geq 1)$, after updating the estimate of λ_{i-1} to be $\hat{\lambda}_{i-1}$, we calculate the value of the deviation $\hat{\lambda}_{i-1} - \lambda_{i-1}$. The evolution of the deviation $\{\hat{\lambda}_{i-1} - \lambda_{i-1}, i \geq 1\}$ is depicted in Fig. 4 (left). The evolution of the a posteriori estimate error variance $\{P_{i-1}, i \geq 1\}$ is depicted in Fig. 4 (right). It is observed that, with the DM number increasing, the deviation and the estimate error variance converge to zero rapidly, showing the competence and

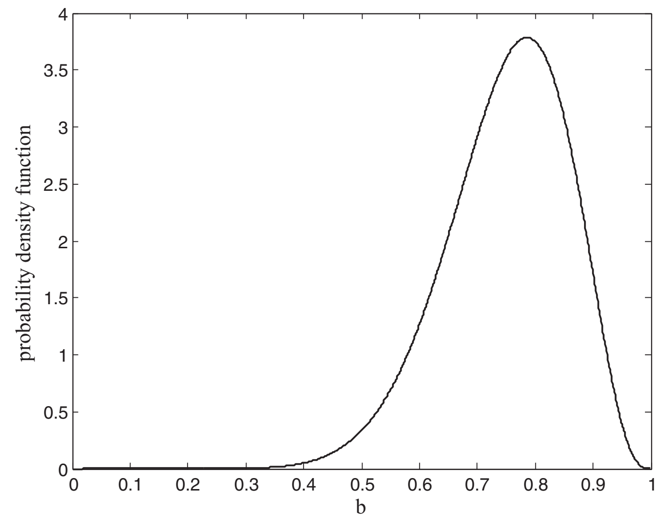

Fig. 2. Probability density function of the degradation-rate-reduction factor.

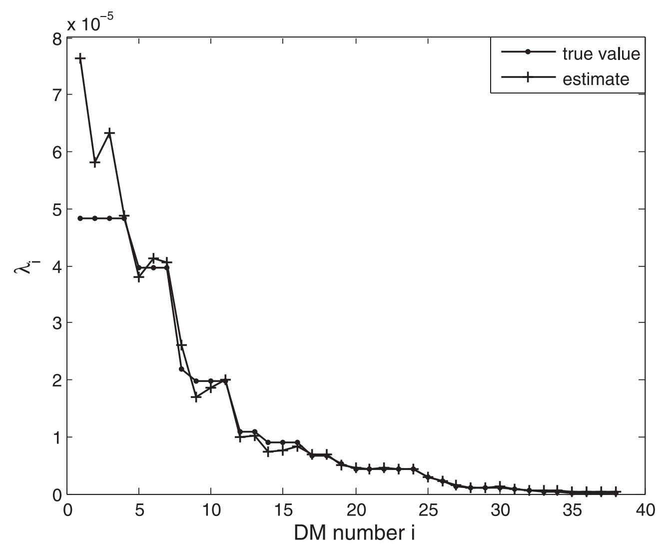

Fig. 3. True values and estimates of the scale parameter.

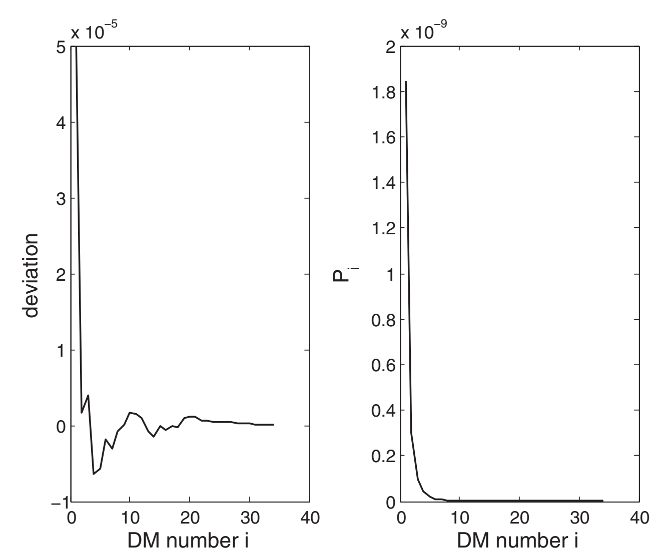

Fig. 4. Evolving paths of $\{\hat{\lambda}_{i-1} - \lambda_{i-1}, i \geq 1\}$ and $\{P_{i-1}, i \geq 1\}$.

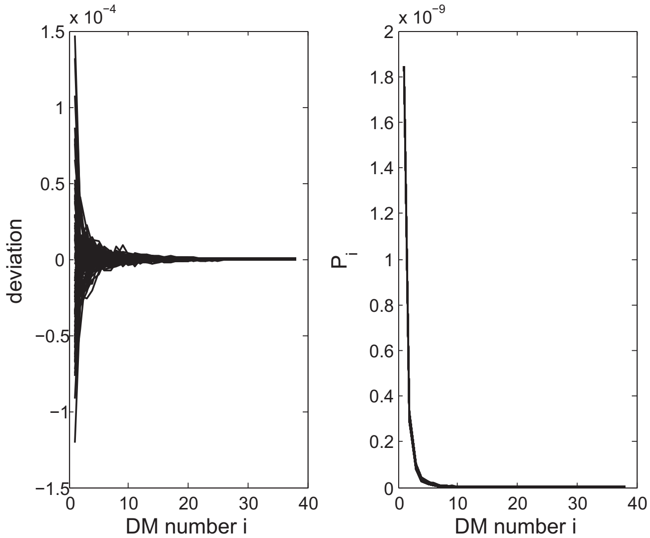

Fig. 5. Evolving paths of $\{\hat{\lambda}_{i-1} - \lambda_{i-1}, i \geq 1\}$ and $\{P_{i-1}, i \geq 1\}$ with 100 samples.

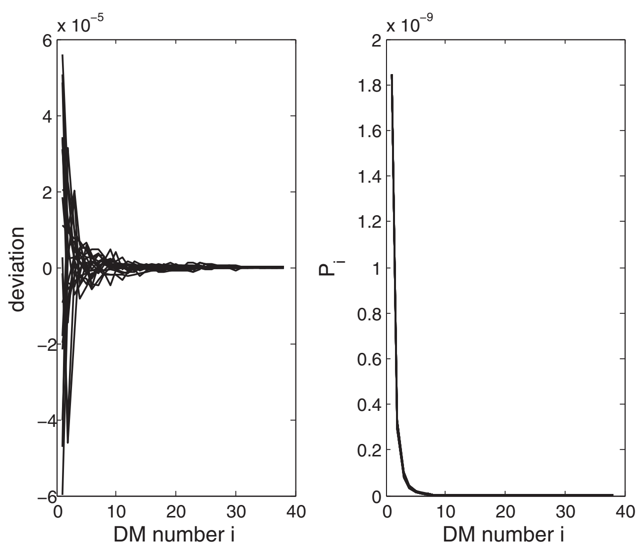

Fig. 6. The impact of the number of repairs on the performance of the Kalman filter.

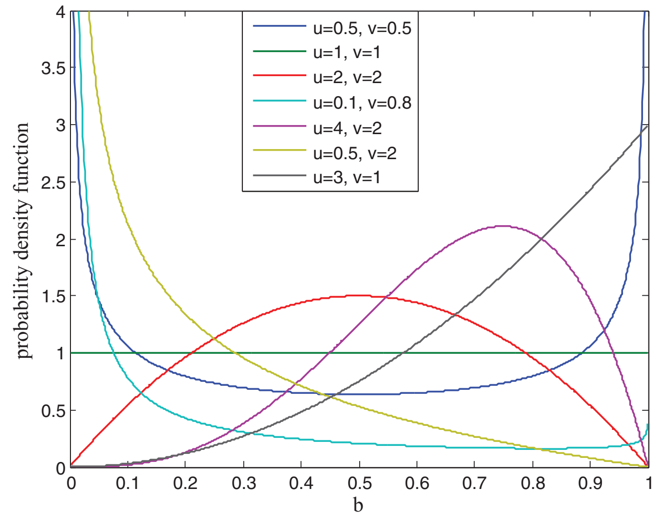

Fig. 7. Different shapes of the beta density function.

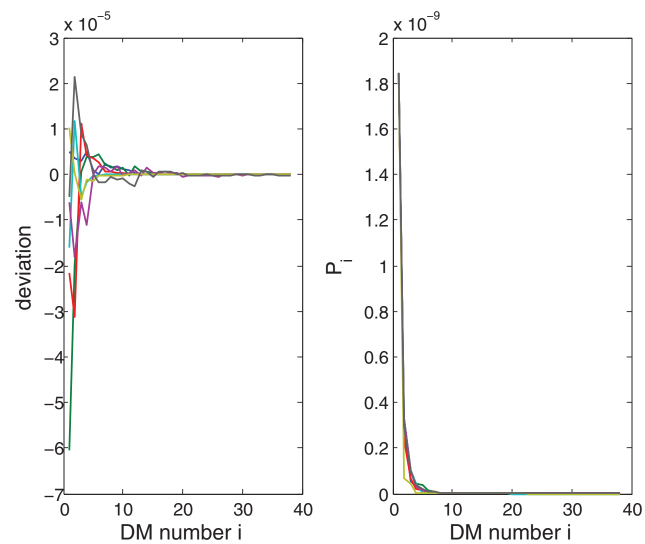

Fig. 8. Evolving paths of $\{\hat{\lambda}_{i-1} - \lambda_{i-1}, i \geq 1\}$ and $\{P_{i-1}, i \geq 1\}$ for different density functions of b.

efficiency of the developed filter. The filter can return a very accurate estimate even when the data size is small.

To show the robustness of the Kalman filter, we run the DG method for 100 times to generate 100 different degradation data sets. Run the recursive filter on the data sets respectively. The 100 evolving paths of the deviation $\{\hat{\lambda}_{i-1} - \lambda_{i-1}, i \geq 1\}$ and 100 evolving paths of the estimate error variance $\{P_{i-1}, i \geq 1\}$ are plotted in Fig. 5. In Fig. 5, all the evolving paths of the deviation and estimate error variance converge rapidly to zero, showing the robustness of the Kalman filter in assessing the scale parameter.

A sensitivity study is conducted to investigate the impact of the number of repairs on the performance of the Kalman filter. Intuitively, with the repairs being more intensive, the scale parameter may be more difficult to estimate. With the values of all the other parameters being the same, we gradually increase the success probability in the Bernoulli distribution from 0.05 to 0.95 with step size 0.05. For each success probability, use the DG method to generate a degradation data set and run the recursive filter on the simulated degradation data. The 19 evolving paths of the deviation $\{\hat{\lambda}_{i-1} - \lambda_{i-1}, i \geq 1\}$ and 19 evolving paths of the estimate error variance $\{P_{i-1}, i \geq 1\}$ are plotted in Fig. 6. Likewise, in Fig. 6, all the evolving paths of the deviation and estimate error variance converge rapidly to zero. For any success probability, the Kalman filter can return an accurate estimate of the scale parameter requiring only about 15 degradation measurements. Therefore, the performance of the proposed Kalman filter is insensitive to the frequency of maintenance.

We further study the impact of the distribution function of the degradation-rate-reduction factor on the performance of the Kalman filter, by varying the values of the two shape parameters u and v. The beta density function can take a great diversity of shapes depending on the values of the two parameters; see Fig. 7. With the values of all the other parameters being the same, we use the DG method to generate a degradation data set for each pair of (u, v) annotated in Fig. 7. Run the recursive filter on the simulated degradation data. The 7 evolving paths of the deviation $\{\hat{\lambda}_{i-1} - \lambda_{i-1}, i \geq 1\}$ and 7 evolving paths of the estimate error variance $\{P_{i-1}, i \geq 1\}$ are plotted in Fig. 8. Fig. 8 shows that the sequences $\{\hat{\lambda}_{i-1} - \lambda_{i-1}, i \geq 1\}$ and $\{P_{i-1}, i \geq 1\}$ converge rapidly to zero. The convergent sequences, $\{\hat{\lambda}_{i-1} - \lambda_{i-1}, i \geq 1\}$ and $\{P_{i-1}, i \geq 1\}$, verify the robustness of the Kalman filter. Therefore, we say that the Kalman filter is insensitive to the starting values, the frequency of maintenance and the distribution function of the degradation-rate-reduction factor. As long as the four parameters,

Table 1
Maximum likelihood estimates of the fixed model parameters ($n = 20$).

$\Pr(m_j = 1)$	0.1	0.2	0.3	0.4	0.5
$\sum_{j=1}^{n-1} m_j$	3	4	6	9	11
$\hat{\lambda}$ (e−5)	9.1041	8.8512	9.5011	9.7829	3.9681
$\hat{\theta}$	2.2904	2.5187	2.6396	2.2955	2.7928
$\hat{\sigma}$	2.4172	2.0156	2.1522	1.7857	2.2023
$\hat{\mu}$	12.9256	11.8338	12.6452	11.3626	9.6170
$\hat{v}$	3.3160	3.7364	3.6416	4.9281	3.7359

Table 2
Maximum likelihood estimates of the fixed model parameters ($n = 50$).

$\Pr(m_j = 1)$	0.1	0.2	0.3	0.4	0.5
$\sum_{j=1}^{n-1} m_j$	5	11	15	20	25
$\hat{\lambda}$ (e−5)	5.1741	7.4122	9.2769	4.4555	3.6644
$\hat{\theta}$	2.7440	2.5159	2.4384	2.6695	2.7430
$\hat{\sigma}$	1.7981	2.0586	2.0659	1.8389	2.4481
$\hat{\mu}$	12. 7328	12.2003	12.3054	11.6542	5.7369
$\hat{v}$	4.2056	4.0101	3.9407	3.3055	4.2793

Table 3
Maximum likelihood estimates of the fixed model parameters ($n = 100$).

$\Pr(m_j = 1)$	0.1	0.2	0.3	0.4	0.5
$\sum_{j=1}^{n-1} m_j$	10	20	29	39	51
$\hat{\lambda}$ (e−5)	6.0332	6.8595	5.3426	9.6861	4.9571
$\hat{\theta}$	2.5392	2.4967	2.7384	2.8622	2.1595
$\hat{\sigma}$	1.9802	2.0272	1.9828	1.9424	1.6610
$\hat{\mu}$	12.3395	12.0103	12.2843	14.2157	8.8489
$\hat{v}$	3.1924	4.0805	5.1244	5.9467	4.5059

$\{\theta, \sigma, u, v\}$, are accurately evaluated, the Kalman filter is able to produce an accurate estimate of the scale parameter after a small number of iterations.

4.2. Performance of the QMC method

The performance of the Kalman filter depends on whether the parameters $\{\theta, \sigma, u, v\}$ are accurately evaluated. Hence, we further study the performance of the QMC method by using simulated degradation data. The true values of the fixed parameters are identical to what have been given in Section 4.1: $\lambda = 7e{-5}$, $\theta = 2.5$, $\sigma = 2$, $u = 12$ and $v = 4$. The length of the DM interval is also 100 units of time, i.e. $\triangle = 100$. To study the effect of sample size on the performance of the QMC method, three levels of sample size are examined: small ($n = 20$), moderate ($n = 50$) and large ($n = 100$). For each fixed sample size n, we gradually increase the success probability in the Bernoulli distribution from 0.1 to 0.5 with step size 0.1. For each pair of sample size and success probability, invoke the DG method to generate a degradation data set and run the QMC method on the simulated degradation data. The ML estimates of $\{\lambda, \theta, \sigma, u, v\}$ (rounded to four decimal places) are summarized in Table 1 (for $n = 20$), in Table 2 (for $n = 50$), and in Table 3 (for $n = 100$).

In Tables 1–3, $\sum_{j=1}^{n-1} m_j$ represents the total number of repairs in the collected data. It can be observed from Table 1 that, with the total number of repairs increasing, the performance of the QMC method first improves and then degrades. Particularly, when the success probability in the Bernoulli distribution is around 0.2, the QMC method can return accurate maximum likelihood estimates for the fixed model parameters. Tables 2 and 3 also demonstrate that with the sample size n fixed and the total number of repairs increasing, the performance of the QMC method first improves and then degrades. However, with the number of repairs increasing, the number of trials N for the Monte Carlo integration shall increase in order to obtain acceptable precision in the answer. Therefore, with the number of repairs increasing, it will take more time for the QMC method to obtain the maximum likelihood estimates.

By comparing Tables 1–3, it is clear that with the total number of repairs fixed and the sample size increasing, the performance of the QMC method improves. For example, when the total number of repairs is 11, the maximum likelihood estimates in Table 2, column 2 are more accurate than those in Table 1, column 5. Similarly, when the total number of repairs is 20, the maximum likelihood estimates in Table 3, column 2 are more accurate than those in Table 2, column 4. Therefore, we say that, with the total number of repairs fixed, the larger the sample size is, the better the performance of the QMC method will be.

Table 4
Drift degradation data from an inertial platform.

Time (hour)	Gyroscopic drift (degree/hour) Item 1	Item 2	Item 3	Item 4	Item 5
2.5	0.0507	0.1300	0.0135	0.0928	0.1052
5.0	0.1789	0.2554	0.0309	0.2633	0.1996
7.5	0.2059	0.3153	0.0077	0.3010	0.2927
10.0	0.2548	0.3966	0.1510	0.3515	0.3060
12.5	0.3117	0.2424	0.1162	0.2818	0.2860
15.0	0.5236	0.3473	0.0237	0.4375	0.2918
17.5	**0.6114**	0.4994	0.2681	0.3357	0.3540
20.0	**0.6616**	**0.7105**	0.4934	0.4413	0.4326
22.5	**1.8166**	**2.6399**	**0.6513**	**0.8157**	0.5042

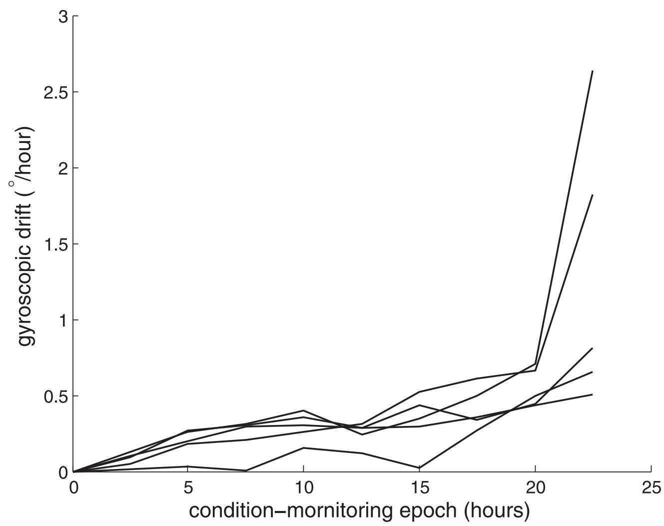

Fig. 9. Drift degradation paths of gyros.

4.3. Real data analysis

For inertial navigation systems, the drift of a gyro is served as a performance indicator in evaluating the physical condition of an inertial platform; see Woodman (2007) for a general description of inertial platforms and gyros. Si et al. (2012) provided a set of drift degradation measurements (along the sense axis). The experimental degradation data from 5 monitored gyros are shown in Table 4. In the experiment, the drift degradation of each gyro was monitored every 2.5 hours. The experiment was terminated after 22.5 hours. It is required that the drift along the sense axis should not exceed 0.6 degree/hour, beyond which the mission of the inertial platform is no longer fulfilled. In Table 4, the drift degradation measurements that exceed 0.6 degree/hour are in bold. The drift degradation paths of the 5 monitored gyros are plotted in Fig. 9. It is clearly observed that the degradation paths are non-monotone and non-linear. Therefore, the non-stationary Wiener process is

Table 5
An optimal maintenance scheme implemented on the inertial platform.

i	1	2	3	4
$EC^i_{pm}(t^*_p)$	2904.9539	1906.5516	1460.2689	1102.9636
$EC^i_{na}(t^*_n)$	2882.6133	1882.6529	1447.1256	1099.7481
m_i	1	1	1	1
t^*_p	6.9695	5.7192	4.9649	4.5850
t^*_n	6.7624	5.3109	4.1406	3.8427
λ_{i-1}	1.3082e−18	9.4790e−19	8.6354e−19	7.4627e−19
$\hat{\lambda}_{i-1}$	3.7707e−10	2.3292e−11	5.8462e−13	4.3576e−15
i	5	6	7	8
$EC^i_{pm}(t^*_p)$	910.6783	753.4435	672.9389	134.4045
$EC^i_{na}(t^*_n)$	908.2469	751.1448	671.3888	138.3805
m_i	1	1	1	0
t^*_p	3.7847	3.4829	2.7738	1.4798
t^*_n	3.6748	3.0543	2.7003	1.3441
λ_{i-1}	5.8059e−19	4.6682e−19	3.7653e−19	2.3675e−19
$\hat{\lambda}_{i-1}$	4.9123e−16	3.0366e−17	1.2388e−17	1.2362e−18

suitable to fit the data set. See Si et al. (2012) for the justification of fitting the drift degradation data by the non-stationary Wiener process with drift function $v(t) = \lambda t^\theta$. Fitting the non-stationary Wiener process $\{X_t = \lambda t^\theta + \sigma W_t\}$ to the drift degradation data, the maximum likelihood estimates of the unknown parameters are obtained as $\hat{\lambda} = 1.3082\text{e}{-18}$, $\hat{\theta} = 13.2024$ and $\hat{\sigma} = 0.1604$. The maximized log-likelihood is -2.1184. If the drift function takes the exponential form: $v(t) = \lambda \exp(\theta t)$. The corresponding maximum likelihood estimates are $\hat{\lambda} = 1.5592\text{e}{-6}$, $\hat{\theta} = 0.5920$ and $\hat{\sigma} = 0.1602$. The maximized log-likelihood is -2.0684. Hence, according to the maximized log-likelihood, the power model, $\{X_t = \lambda t^\theta + \sigma W_t\}$, is practically equal to the exponential model, $\{X_t = \lambda \exp(\theta t) + \sigma W_t\}$.

We then implement the proposed maintenance policy on the inertial platforms to illustrate the procedure for determining optimal maintenance actions. At each DM point, having obtained the drift degradation measurement, maintenance technicians have to decide whether or not to preventively maintain the target inertial platform. Preventive maintenance on the inertial platform, e.g., replacing one of the gyros, is assumed to be imperfect. Unexpected failure of the inertial platform is hazardous and costly. The inertial platform itself is also very expensive. A high-end marine-grade inertial navigation system may cost over 1 million dollars. In the present paper we consider an industrial-grade inertial navigation system, and the cost configurations are specified as follows. The cost of replacing an inertial platform is $c_r = \$2000$. The cost of an unexpected failure of the inertial platform is $c_f = \$10\,000$. The cost of each inspection is $c_I = \$20$. A PM action on the inertial platform costs $c_p = \$100$. Here, c_p need not be constant during a whole replacement cycle, but can vary according to practical conditions.

Because there was no maintenance in the experiment, we use the DG method to simulate degradation data and maintenance data. The parameters of the non-stationary Wiener process assume the values estimated: $\lambda = 1.3082\text{e}{-18}$, $\theta = 13.2024$ and $\sigma = 0.1604$. To be consistent, the two shape parameters assume the same values as what have been assigned in Section 4.1: $u = 12$ and $v = 4$. Fit the non-stationary Wiener process $\{X_t = \lambda t^\theta + \sigma W_t\}$ to the drift degradation data of each gyro, and calculate the variance of the five estimates of the scale parameter. The starting value, P_0, is assigned with the variance of the five estimates: $P_0 = 3.5995\text{e}{-11}$. The starting value, $\hat{\lambda}_0$, is assigned with the estimate of the scale parameter obtained by fitting the non-stationary Wiener process to the drift degradation data of the first four gyros: $\hat{\lambda}_0 = 1.3431\text{e}{-18}$. Once a new degradation measurement is obtained, we immediately update the a posteriori estimate of the scale parameter and the a posteriori estimate error variance. Under the preceding parameter settings and cost configurations, one realization of the replacement cycle is described in Table 5.

Table 6
Optimal preventive-replacement times and the corresponding minimal expected cost rates—calculated based on the λ_{i-1}'s in Table 5.

i	1	2	3	4
$EC^i_{pm}(t^*_p)$	190.7614	185.7332	181.6311	175.6586
$EC^i_{na}(t^*_n)$	186.2923	181.4898	176.2722	171.9305
t^*_p	7.4776	7.0293	6.8275	6.6115
t^*_n	6.9776	6.5986	6.4756	5.9054
i	5	6	7	8
$EC^i_{pm}(t^*_p)$	149.5069	137.2797	125.3346	120.1797
$EC^i_{na}(t^*_n)$	146.6490	135.2280	124.6904	121.8193
t^*_p	6.4081	5.9653	5.7070	4.7836
t^*_n	5.8409	5.6314	5.3150	4.3794

According to Table 5, the decision-making process involves the following steps. At time $\triangle$, since the minimized cost rate $EC^1_{pm}(t^*_p)$ is smaller than its counterpart $EC^1_{na}(t^*_n)$, the system is preventively maintained at time $t = \triangle$. Since the optimal preventive-replacement time t^*_p is larger than the DM interval, the drift degradation is monitored at the second DM point. Likewise, at the ensuing DM point $2\triangle$, the optimal maintenance decision is to measure the drift degradation at time $3\triangle$ with a repair performed at time $2\triangle$. At time $8\triangle$, since the minimized cost rate $EC^8_{pm}(t^*_p)$ is larger than its counterpart $EC^8_{na}(t^*_n)$, the system is left intact. Since the optimal preventive-replacement time t^*_n is smaller than the DM interval, the system is preventively replaced at time $t = 8\triangle + 1.3441$, if it survives to this point. Table 5 also lists the true value and estimate of the scale parameter at each DM point. Though the estimates are not accurate, it can be seen that the estimation errors $\{\hat{\lambda}_{i-1} - \lambda_{i-1}, i = 1, \ldots, 8\}$ converge to zero.

Table 6 lists the optimal preventive-replacement times and the corresponding minimal expected cost rates that are calculated by using the true values λ_{i-1}'s. By comparing Tables 5 and 6, it can be seen that the sequences of optimal preventive-replacement times and the sequences of minimal expected cost rates show the same evolving trend. According to Table 6, the system will also be maintained at time points $i\triangle$ for $i = 1, \ldots, 7$. The difference is that, at time $8\triangle$, the system is left intact but subject to an inspection at time $9\triangle$.

5. Conclusions

In this paper, we developed an imperfect maintenance model for systems whose sensor information can be modeled by stochastic processes. There are three main contributions of this paper. The first contribution is the modeling of imperfect maintenance via the degradation rate function instead of the hazard rate function. Our model is more suitable and practical for degradation-based maintenance. The second contribution is the introduction of a random improvement factor. The assumption of random improvement factors is more realistic than the assumption of constant improvement factors. The third contribution is the development of two parameter-estimation methods for estimating different types of unknown parameters in the model. Numerical studies and sensitivity analysis have verified the feasibility, robustness and competence of the parameter-estimation methods. The model and numerical study presented here have left a number of practical issues untouched. Many of these can be dealt with by appropriate technical extensions.

- As mentioned before, the extended improvement factor model involves the degradation-rate-reduction factor and the shifting factor, yet not the age-reduction factor. Future work can be done to develop a model with three improvement factors.
- The non-stationary Wiener process represents a family of continuous yet non-monotone diffusion processes. Other stochastic processes, such as non-stationary Gamma processes and non-stationary inverse Gaussian processes, can be used to characterize monotonically drifted diffusion processes. The imperfect maintenance model and the two parameter-estimation methods can be extended to monotonically drifted diffusion processes by appropriate modifications.
- Numerical examples showed that the QMC method is efficient when the total number of repairs is relatively small (compared with the size of the collected data). However, it will fail when the total number of repairs is relatively large. Therefore, it is of practical need to develop more competent methods for estimating fixed model parameters.

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Appendix A. Updating remaining useful life distribution

For any $i \geq 1$, after obtaining the degradation datum x_i ($< l$), we immediately update the estimate of the scale parameter to be $\hat{\lambda}_{i-1}$. From time $i\triangle$ forward, the degradation of the system can be modeled by $\{X_t = \mu_i\hat{\lambda}_{i-1}t^{\theta} - \mu_i\hat{\lambda}_{i-1}(i\triangle)^{\theta} + x_i + \sigma W_{t-i\triangle}, t > i\triangle\}$. By definition, the lifetime of the system is the first hitting time of the process $\{X_t, t > i\triangle\}$ to the critical threshold l. Define a non-negative function $\upsilon_i(t)$:

$$\upsilon_i(t) = \mu_i\hat{\lambda}_{i-1}(t+i\triangle)^{\theta} - \mu_i\hat{\lambda}_{i-1}(i\triangle)^{\theta}.$$

We have $\upsilon_i(0) = 0$. Define a modified critical threshold l_i:

$$l_i = l - x_i.$$

It can be verified that the remaining useful life of the system is the first hitting time of the process $\{Y_t^i = \upsilon_i(t) + \sigma W_t, t \geq 0\}$ to the threshold l_i.

For a non-stationary Wiener process, deriving the first hitting time distribution involves solving the Fokker–Planck–Kolmogorov equation with boundary constraints (Cox & Miller, 1965). Many works, such as Gebraeel, Lawley, Li, and Ryan (2005), Park and Padgett (2005) and Wang (2010), made non-linear transformations on degradation data and fitted the stationary Wiener process to the transformed data. An implicit assumption of the data-transformation approach is that the random part of the transformed process is still Brownian motion, which may not always be the case. Inspired by Si et al. (2012), we developed below an explicit form for approximating the first hitting time distribution function.

According to Si et al. (2012, Corollary 1), the density function of the first hitting time of the process $\{Y_t^i = \upsilon_i(t) + \sigma W_t, t \geq 0\}$ to the threshold l_i can be approximated by

$$\tilde{f}_i(t) = \frac{l_i - \upsilon_i(t) + \theta\mu_i\hat{\lambda}_{i-1}t(t+i\triangle)^{\theta-1}}{\sigma\sqrt{2\pi t^3}} \exp\left(-\frac{[l_i - \upsilon_i(t)]^2}{2\sigma^2 t}\right), \quad t > 0.$$

However, it can be verified that the integral of $\tilde{f}_i(t)$ over the interval $(0, +\infty)$ does not equal one. Therefore, in order to make it a probability density function, we multiply $\tilde{f}_i(t)$ by a normalizing constant. Consequently, we can state the following result.

Proposition 1. For any $i \geq 1$, the remaining useful lifetime distribution function $P_{rul}(t|\ddot{X}_i, \ddot{M}_i)$ can be approximated by

$$\tilde{F}_i(t) = \varphi_i \times \int_0^t \frac{l_i - \upsilon_i(s) + \theta\mu_i\hat{\lambda}_{i-1}s(s+i\triangle)^{\theta-1}}{\sigma\sqrt{2\pi s^3}} \times \exp\left(-\frac{[l_i - \upsilon_i(s)]^2}{2\sigma^2 s}\right) ds, \quad t > 0.$$

Here, φ_i is a normalizing constant:

$$\frac{1}{\varphi_i} = \int_0^{\infty} \frac{l_i - \upsilon_i(s) + \theta\mu_i\hat{\lambda}_{i-1}s(s+i\triangle)^{\theta-1}}{\sigma\sqrt{2\pi s^3}} \exp\left(-\frac{[l_i - \upsilon_i(s)]^2}{2\sigma^2 s}\right) ds.$$

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